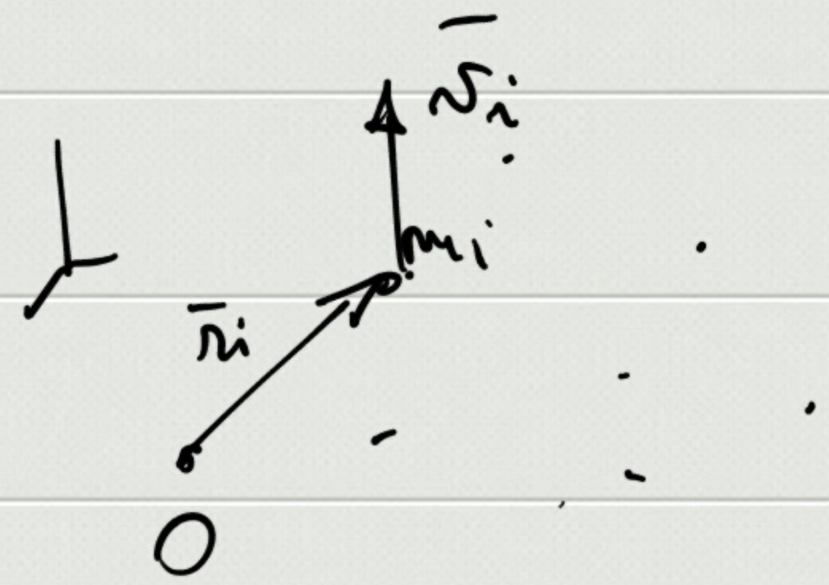
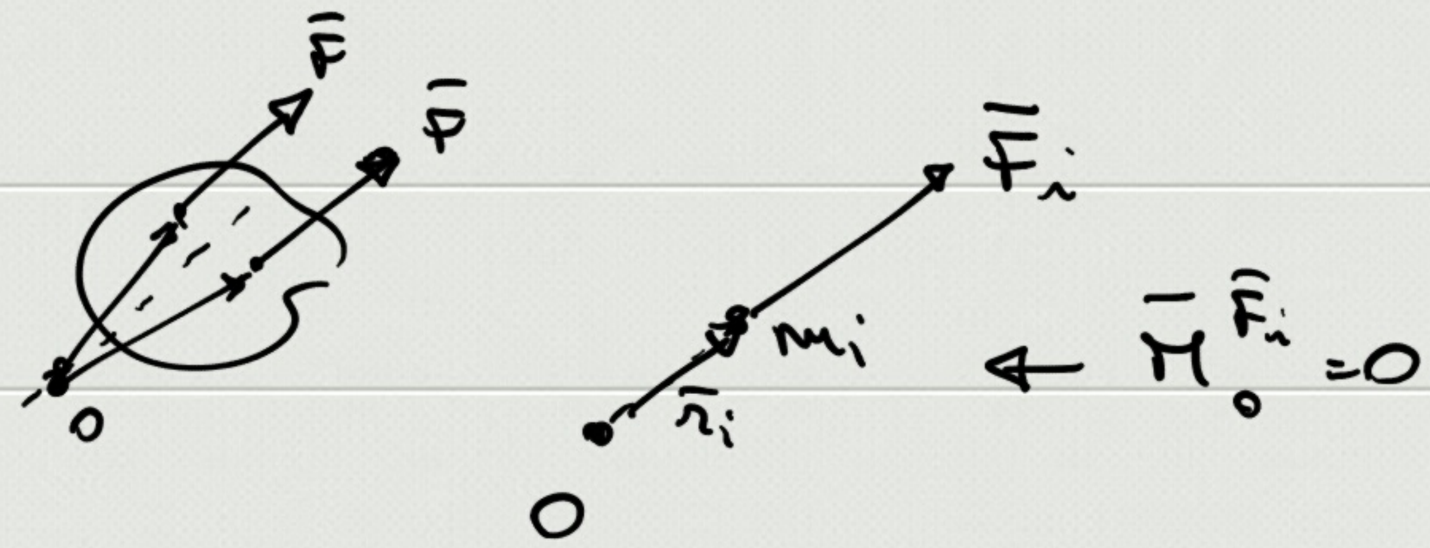
$$\frac{d\bar{L}_O}{dt} = -\bar{v}_O \times \cancel{m_{tot}\,\bar{v}_{CM}} + \bar{M}_O^E$$

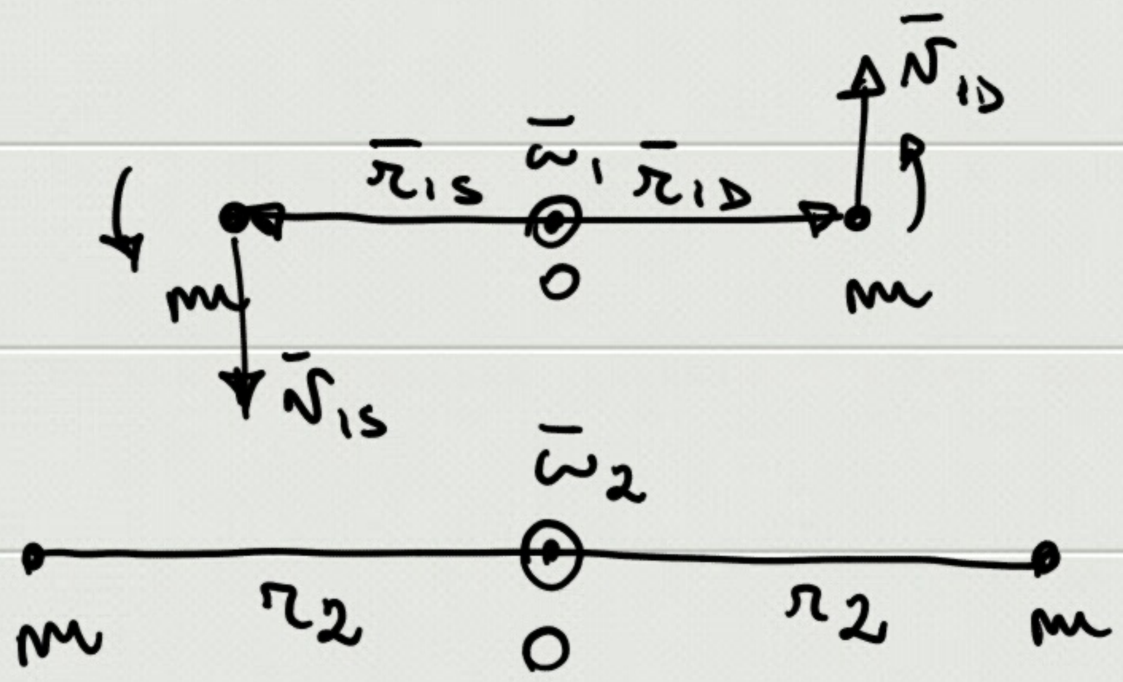


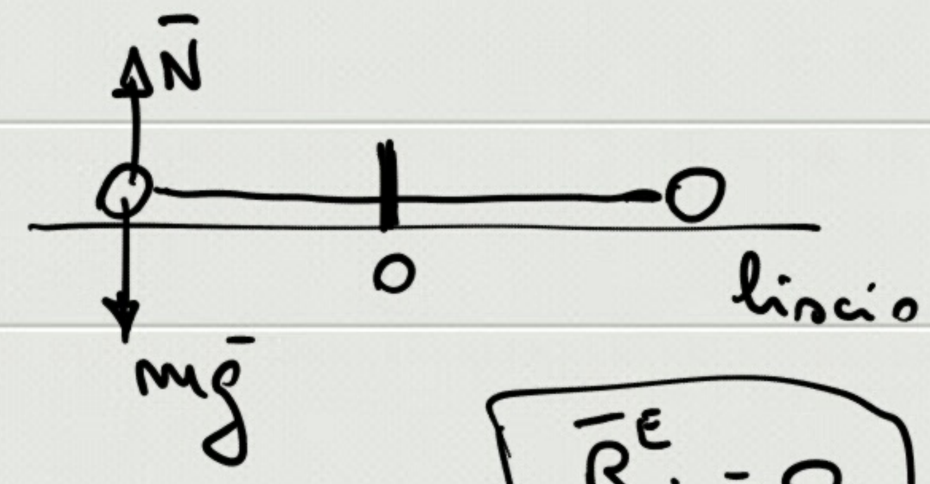
$\bar{M}_O^E = 0$ $\qquad$ $\bar{R}_i = 0$ $\Rightarrow$ sistema isolato



$$\Rightarrow \quad \frac{d\bar{L}_O}{dt} = \bar{M}_O^E = 0 \quad \Rightarrow \quad \boxed{\bar{L}_O = \text{cost}}$$

$$\frac{dL_{O,\alpha}}{dt} = M^E_{O,\alpha} = 0 \quad \Rightarrow \quad \boxed{L_{O,\alpha} = \text{cost}}$$





$$\boxed{\bar{R}^E_i = 0}$$

Polo: O $\qquad \bar{L}_{O,1} = \bar{r}_{1D} \times m\bar{v}_{1D} + \bar{r}_{1S} \times m\bar{v}_{1S} =$

$$= 2\,\bar{r}_1 \times m\bar{v}_1 \qquad \odot\,\bar{L}$$

Sist. isolato $\Rightarrow \boxed{\bar{L}_O = \text{cost}} \Rightarrow \bar{L}_{O,1} = \bar{L}_{O,2}$

$$\Rightarrow L_{O,1} = L_{O,2}$$

$$|\bar{L}_O| = 2\,r\,m\,v = 2\,m\,\omega\,r^2$$

$$\Rightarrow \cancel{2m}\,\omega_1 r_1^2 = \cancel{2m}\,\omega_2 r_2^2$$

$$\Rightarrow \omega_2 = \omega_1 \left(\frac{r_1}{r_2}\right)^2 < \omega_1$$